Introduction to Supervised Learning

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1. Supervised Learning
2. Statistical Learning Theory

1 Supervised Learning

2 Statistical Learning Theory

Definition

Given observations $(x_i, y_i) \in \mathcal{X}, \mathcal{Y}$ of inputs and outputs, predict a new $y \in \mathcal{Y}$ given a new previous unseen $x \in \mathcal{X}$

Potential Difficulties

- y may not be deterministic of x: $y = f(x) + \epsilon$
- f may be complex (f: prediction function)
- only a few x's are observed
- input space $\mathcal{X}$ may be large $\rightarrow$ curse of dimensionality
- may be a weak link between training and testing distribution (distribution shift?)
- choose of criterion of performance

Curse of dimentionality

- Hughes phenomenon（休斯现象）
- distance function: comparasion of (Euclidean) distance will become invalid
  [Beyer et al. 99][1] Fix $\epsilon > 0$ and N. If data is "truly high-dimensional", then under fairly weak assumption on the distribution of the data:

$$\lim_{D\to\infty} P[d_{max}(N, D) \leq (1+\epsilon) d_{min}(N, D)] = 1$$

  where N is the data size, and D is the dimension.
- consistency: slow rate of convergence to bayes risk
- overfitting / underfitting

[1] Beyer K., Goldstein J., Ramakrishnan R., Shaft U. (1999) When Is "Nearest Neighbor" Meaningful?. In: Beeri C., Buneman P. (eds) Database Theory —ICDT' 99. ICDT 1999. Lecture Notes in Computer Science, vol 1540. Springer, Berlin, Heidelberg. https://doi.org/10.1007/3-540-49257-7_15

Performance Evaluation: Loss function

- Loss function: $l(y, z)$ is the loss of predicting z while the true label is y
  - Binary: $1_{y \neq z}$ 0-1 loss
  - Multi-category: $1_{y \neq z}$
  - Regression: $(y - z)^2$ square loss $\quad |y - z|$ absolute loss
  - Structured prediction: $y \in \{0, 1\}^k$ Hamming loss: $\sum_{j=1}^{k} 1_{y_j \neq z_j}$ Hamming Distance: 信息论中，表示两个等长字符串在对应位置上不同字符的数目
- Expected risk: $\mathcal{R}(f) = E[l(y, f(x))] = \int_{\mathcal{X} \times \mathcal{Y}} l(y, f(x)) dp(x, y)$
- Empirical risk: $\hat{\mathcal{R}}(f) = \frac{1}{n} \sum_{i=1}^{n} l(y_i, f(x_i))$
  both are equal in expectation (over the randomness of the training set)

Performance Evaluation: ROC curve I

- ROC curve: Receiver Operating Characteristic Curve
- Confusion Matrix
- 真阳性率：$TPR = \frac{TP}{TP+FN}$ 所有阳性中被正确归为阳性
  伪阳性率：$FPR = \frac{FP}{FP+TN}$ 所有阴性中被错误判为阳性
- 完美的预测：(FPR, TPR) = (0,1)：无伪阳性，都是真阳性
  随机猜：位于直线 $y = x$ 上

		真实值		总数
		p	n	
预测输出	p'	真阳性 (TP)	伪阳性 (FP)	P'
	n'	伪阴性 (FN)	真阴性 (TN)	N'
总数		P	N	

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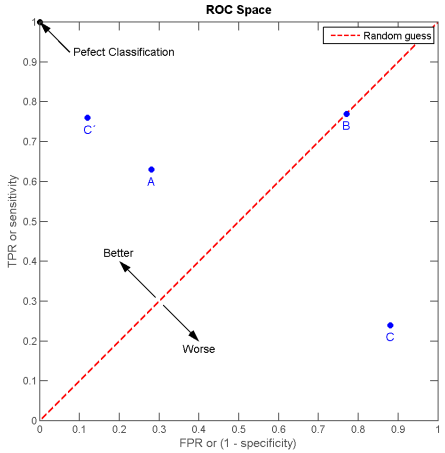
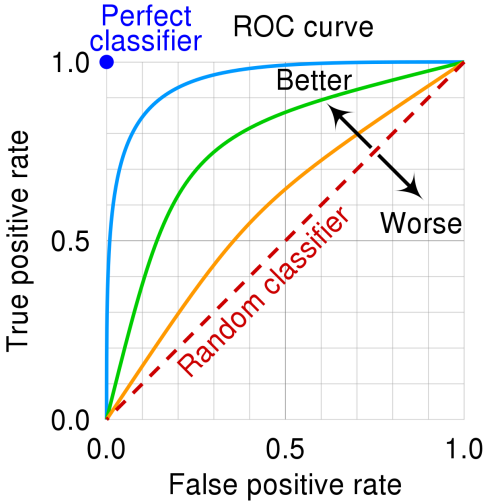
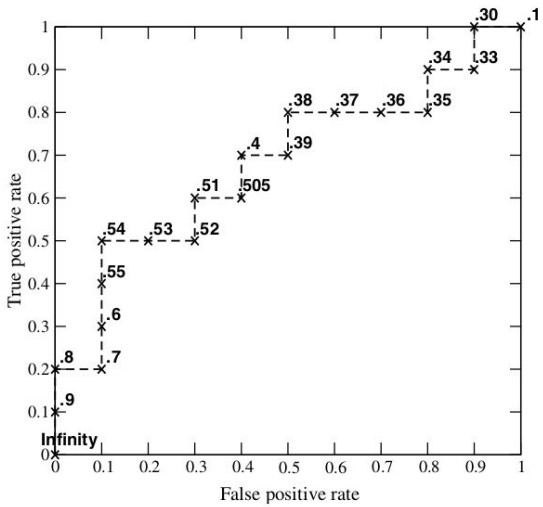




Performance Evaluation: ROC curve II

- ROC 曲线：对于一个二元分类模型，将同一模型每个阈值的（FPR,TPR）坐标都画在 ROC 空间里
- AUC (Area Unver Curve)：ROC 曲线下与坐标轴围成的面积（越接近 1，检验方法真实性越高，等于 0.5 时，真实性最低，无应用价值）
- An example[2]

Inst#	Class	Score	Inst#	Class	Score
1	p	.9	11	p	.4
2	p	.8	12	n	.39
3	n	.7	13	p	.38
4	p	.6	14	n	.37
5	p	.55	15	n	.36
6	p	.54	16	n	.35
7	n	.53	17	p	.34
8	n	.52	18	n	.33
9	p	.51	19	p	.30
10	n	.505	20	n	.1



Score 从高到低，依次计算得到 (FPR, TPR)，连线。

[2] https://www.zhihu.com/question/22844912

Bayes predictor I: What is the best prediction function f?

$$\mathcal{R}(f) = \mathbb{E}\Big[\ell(y, f(x))\Big] = \mathbb{E}\big[\mathbb{E}(\ell(y, f(x))|x)\big],$$

which we can rewrite

$$\mathcal{R}(f) = \mathbb{E}_{x' \sim dp(x')}\big[\mathbb{E}(\ell(y, f(x))|x = x')\big] = \int_{\mathcal{X}} \Big(\mathbb{E}(\ell(y, f(x))|x = x')\Big) dp(x').$$

Given the conditional distribution given any $x' \in \mathcal{X}$, that is $y|x = x'$, we can define the *conditional risk* for any $z \in \mathcal{Y}$ (it is a deterministic function):

$$r(z|x') = \mathbb{E}(\ell(y, z)|x = x'),$$

which leads to

$$\mathcal{R}(f) = \mathbb{E}(r(f(x)|x)) = \mathbb{E}_{x' \sim dp(x')}\big[r(f(x')|x')\big] = \int_{\mathcal{X}} r(f(x')|x') dp(x').$$

A minimizer of $\mathcal{R}(f)$ can be obtained by considering for any $x' \in \mathcal{X}$, the function value $f(x')$ to be equal to a minimizer $z \in \mathcal{Y}$ of $r(z|x') = \mathbb{E}(\ell(y, z)|x = x')$. We can therefore **consider all x' as being treated independently**. This leads to the following propositions.

Proposition 2.1 (Bayes predictor and Bayes risk) *The expected risk is minimized at a* Bayes predictor $f^* : \mathcal{X} \to \mathcal{Y}$ *satisfying for all* $x' \in \mathcal{X}$, $f^*(x') \in \arg\min_{z \in \mathcal{Y}} \mathbb{E}(\ell(y, z)|x = x') = \arg\min_{z \in \mathcal{Y}} r(z|x')$. *The Bayes risk* $\mathcal{R}^*$ *is the risk of all Bayes predictors and is equal to*

$$\mathcal{R}^* = \mathbb{E}_{x' \sim dp_x(x')} \inf_{z \in \mathcal{Y}} \mathbb{E}(\ell(y, z)|x = x').$$

Bayes predictor II

Definition 2.3 (Excess risk) *The excess risk of a function $f : \mathcal{X} \to \mathcal{Y}$ is equal to $\mathcal{R}(f) - \mathcal{R}^*$ (it is always non-negative).*

- **Binary classification**: the Bayes predictor for $\mathcal{Y} = \{0, 1\}$ and $\ell(y, z) = 1_{y \neq z}$ is such that

  $$f^*(x') \in \arg\min_{z \in \{0,1\}} \mathbb{P}(y \neq z | x = x') = \arg\min_{z \in \{0,1\}} 1 - \mathbb{P}(y = z | x = x') = \arg\max_{z \in \{0,1\}} \mathbb{P}(y = z | x = x').$$

  The optimal classifier will select the most likely class given x'.

- **Regression**: the Bayes predictor for $\mathcal{Y} = \mathbb{R}$ and $\ell(y, z) = (y - z)^2$ is such that

  $$f^*(x') \in \arg\min_{z \in \mathbb{R}} \mathbb{E}\big[(y - z)^2 | x = x'\big] = \arg\min_{z \in \mathbb{R}} \Big\{\mathbb{E}\big[(y - \mathbb{E}(y|x = x'))^2 | x = x'\big] + (z - \mathbb{E}(y|x = x'))^2\Big\}.$$

  This leads to the conditional expectation $f^*(x') = \mathbb{E}(y|x = x')$.

Exercise 2.1 *What is the Bayes predictor for regression with the absolute loss $\ell(y, z) = |y - z|$?*

Learning from data (to get f)

- Local averaging. eg. KNN
- ERM (Empirical Risk Minimization)

Consider a parameterized family of prediction functions $f_\theta : \mathcal{X} \to \mathcal{Y}$ for $\theta \in \Theta$ and minimize the empirical risk with respect to $\theta \in \Theta$:

$$\hat{\mathcal{R}}(f_\theta) = \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, f_\theta(x_i)).$$

The most classical example is linear least-squares regression (studied at length in Chapter 3), where we minimize

$$\frac{1}{n} \sum_{i=1}^{n} (y_i - \theta^\top \varphi(x_i))^2,$$

- Risk decomposition in estimation error + approximation error : given any $\hat{\theta} \in \Theta$, we can write the excess risk of $f_{\hat{\theta}}$ as:

$$\begin{aligned} \mathcal{R}(f_{\hat{\theta}}) - \mathcal{R}^* &= \left\{ \mathcal{R}(f_{\hat{\theta}}) - \inf_{\theta' \in \Theta} \mathcal{R}(f_{\theta'}) \right\} + \left\{ \inf_{\theta' \in \Theta} \mathcal{R}(f_{\theta'}) - \mathcal{R}^* \right\} \\ &= \text{estimation error} \quad + \text{approximation error} \end{aligned}$$

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2 Statistical Learning Theory

Machine Learning v.s. Statistical Learning

- Machine Learning: Algorithms, Computational Scalability
- Statistical Learning: Models, Interpretability

Goal of Statistical Learning Theory

- Goal: provide some guarantees of performance on unseen data
- Assumption: i.i.d. data, $\mathcal{D}_n(dp) = \{(x_1, y_1), \cdots, (x_n, y_n)\}$
- Algorithm: a mapping from $\mathcal{D}_n(dp)$ to a function from $\mathcal{X}$ to $\mathcal{Y}$
- Risk: $\mathcal{R}_{dp}(f)$
- Goal: min $\mathcal{R}_{dp}(\mathcal{A}(\mathcal{D}_n(dp))) - \mathcal{R}^*_{dp}$

Measure of performance

- Expected Error: $E[\mathcal{R}_{dp}(\mathcal{A}(\mathcal{D}_n(dp)))]$
  Consistency: $E[\mathcal{R}_{dp}(\mathcal{A}(\mathcal{D}_n(dp))) - \mathcal{R}^*_{dp}]$ goes to 0 as $n \to \infty$
- PAC Learning: (Probably approximately correct)
  For a given $\delta \in (0, 1)$ and $\epsilon > 0$:

$$P([\mathcal{R}_{dp}(\mathcal{A}(\mathcal{D}_n(dp))) - \mathcal{R}^*_{dp}] \leq \epsilon) \geq 1 - \delta$$

  Find $\epsilon(\delta)$ which is as small as possible
  Consistency: δ_n tends to 0 for any $\epsilon > 0$

Thanks!